

Slope and Rate of Change — Diagnosing Inverted and Sign-Reversed Calculations

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

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|------------------|--------|---------|
| 1. 2 | 4. -18 | 7. -1.5 |
| 2. -2 | 5. -3 | 8. 5 |
| 3. $\frac{1}{2}$ | 6. 2 | |
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Curriculum

Level: Grade 8 / Pre-Algebra

8.EE.B.5, 8.EE.B.6 – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–5 of 5 across 8 tagged checks.

Common wrong answers

1. If they got 0.5: inverted the ratio: run over rise instead of rise over run
 2. If they got 2: subtracted the y -values in one order and the x -values in the other, reversing the sign
 3. If they got 2: divided the change in x by the change in y , inverting the slope
 4. If they got 18: subtracted the smaller reading from the larger one, losing the sign of the change
 5. If they got 3: subtracted the levels in one order and the times in the other
 6. If they got -2: took the y -difference from the first point to the second but the x -difference the other way
 7. If they got -0.67: inverted the ratio, dividing the change in time by the change in height
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1. Slope through (2, 3) and (6, 11).

Step 1: label the points in one fixed order: $(x_1, y_1) = (2, 3)$ and $(x_2, y_2) = (6, 11)$. Step 2: rise first (it belongs on top): $y_2 - y_1 = 11 - 3 = 8$. Step 3: run underneath, subtracted in the SAME order: $x_2 - x_1 = 6 - 2 = 4$. Step 4: $m = \frac{8}{4}$, so $\boxed{m = 2}$. Sanity check: from (2, 3) the line climbs 8 while moving 4 right, so a slope greater than 1 is right.

Watch for: 0.5 — the run divided by the rise.

2. Find and fix: Devi's slope through $(-3, 5)$ and $(2, -5)$.

Step 1: read her set-up as an order question. On top she wrote $5 - (-5)$: that is $y_1 - y_2$, first point minus second. Underneath she wrote $2 - (-3)$: that is $x_2 - x_1$, second point minus first. Step 2: name the error — the two differences were taken in *opposite* orders, which multiplies the answer by -1 . That is a sign reversal, not an arithmetic slip; her 10 and her 5 are both correct numbers. Step 3: redo it in one order, second minus first throughout: $m = \frac{-5 - 5}{2 - (-3)} = \frac{-10}{5}$. Step 4: $m = -2$. Check the direction: y falls from 5 to -5 while x increases, so the slope must be negative — Devi's $+2$ could have been rejected before any arithmetic.

3. Find and fix: Mika's slope through $(1, 4)$ and $(5, 6)$.

Step 1: identify what each of his differences measures. His numerator $5 - 1$ is the change in x , and his denominator $6 - 4$ is the change in y : the fraction is upside down. Step 2: name the error — inverted ratio (run over rise). His answer is the *reciprocal* of the true slope, which is why it is 2 instead of a number below 1. Step 3: rebuild it as rise over run: $m = \frac{6 - 4}{5 - 1} = \frac{2}{4}$. Step 4: simplify: $m = \frac{1}{2}$. Check: moving 4 right lifts the line only 2, so the line is shallow and the slope must be less than 1.

4. Change in the tank's level from hour 0 to hour 6.

Step 1: read both values off the table: at hour 0 the level is 46 litres, and at hour 6 it is 28 litres. Step 2: a change is always *later minus earlier* — the order carries the meaning, so it is $28 - 46$, not $46 - 28$. Step 3: $28 - 46 = -18$ litres. Step 4: check the sign against the story: the tank is draining, so the change must be negative.

Watch for: 18 — the same size, but it describes a tank filling up.

5. Rate of change of the tank's level, hour 0 to hour 6.

Step 1: a rate of change is the slope of the table: change in litres over change in hours, both taken later-minus-earlier. Step 2: numerator (from problem 4): $28 - 46 = -18$ litres. Step 3: denominator, same order: $6 - 0 = 6$ hours. Step 4: $\frac{-18}{6} = -3$, that is, 3 litres lost per hour. Step 5: check with a single step of the table: from hour 0 to hour 2 the level drops from 46 to 40, which is 6 litres in 2 hours — the same -3 per hour.

Watch for: 3 — correct size, wrong sign, and it claims the tank is filling.

6. Find and fix: Ty's slope through $(5, 9)$ and $(1, 1)$.

Step 1: check the order of each difference. Ty's numerator $1 - 9$ runs from the first point's y to the second point's y ; his denominator $5 - 1$ runs the other way, second to first. Step 2: name the error — mixed order, so his answer carries the wrong sign. Step 3: fix the order and keep it: taking $(5, 9)$ as point 1, $m = \frac{1 - 9}{1 - 5} = \frac{-8}{-4}$. Step 4: two negatives divide to a positive: $m = 2$. Step 5: direction — a positive slope means the line **ris**es from left to right. Check it against the points: as x grows from 1 to 5, y grows from 1 to 9.

Watch for: -2 — Ty's answer; the size is right, so only the sign diagnosis separates a correct paper from his.

7. Candle: 18 cm at 3 minutes, 6 cm at 11 minutes. Diagnose Jo's rate and find the correct one.

Step 1: read Jo's fraction. She put the time difference $11 - 3$ on top and the height difference $6 - 18$ underneath — minutes per centimetre, the reciprocal of what was asked. Her orders were consistent, so her only error is the inversion. Step 2: set it up as height change over time change, later minus earlier in both: $\frac{6 - 18}{11 - 3}$. Step 3: numerator $6 - 18 = -12$ cm; denominator $11 - 3 = 8$ minutes. Step 4: $\frac{-12}{8} = \boxed{-1.5}$, that is, the candle loses 1.5 cm each minute. Step 5: check plausibility: in 8 minutes it should lose $8 \times 1.5 = 12$ cm, and it does (18 down to 6).

Watch for: -0.67 — Jo's inverted value; note it is between -1 and 0 , far too slow for a candle that lost 12 cm in 8 minutes.

8. Challenge: the line through $(a, 4)$ and $(7, -2)$ has slope -3 . Diagnose Priya's set-up and solve for a .

Step 1: diagnose. Priya wrote $\frac{7 - a}{-2 - 4}$: the x -difference is on top and the y -difference underneath, so her equation says run over rise $= -3$. It is the inverted slope, set equal to the true slope. Step 2: write the correct equation, rise over run, both differences second minus first: $\frac{-2 - 4}{7 - a} = -3$. Step 3: simplify the numerator: $\frac{-6}{7 - a} = -3$. Step 4: multiply both sides by $(7 - a)$: $-6 = -3(7 - a) = -21 + 3a$. Step 5: add 21 to both sides: $15 = 3a$, so $\boxed{a = 5}$. Step 6: check by recomputing the slope with $a = 5$: through $(5, 4)$ and $(7, -2)$, $m = \frac{-2 - 4}{7 - 5} = \frac{-6}{2} = -3$ ✓. Priya's -11 fails this check: through $(-11, 4)$ and $(7, -2)$ the slope is $\frac{-2 - 4}{7 - (-11)} = \frac{-6}{18} = -\frac{1}{3}$, the reciprocal of -3 — the signature of an inverted set-up.